

RALSIN TABLET

DESCRIPTION:

RALSIN TABLET 2 MG: A blue, round tablet (diameter 7 mm) with 'DUOMA/DUOMA' & SCORED marking.

RALSIN TABLET 5 MG: A green, round tablet (diameter 7 mm) with 'DUOMA/DUOMA' & SCORED marking.

COMPOSITION:

RALSIN TABLET 2 MG: Each tablet contains Terazosin hydrochloride dihydrate equivalent to Terazosin 2 mg.

RALSIN TABLET 5 MG: Each tablet contains Terazosin hydrochloride dihydrate equivalent to Terazosin 5 mg.

PHARMACODYNAMICS:

Terazosin hydrochloride is an α -1 selective adrenoceptor blocking agent. Blockade of α -1 adrenergic receptors inhibits vasoconstriction induced by endogenous catecholamines; vasodilation may occur in both arteriolar resistance vessels and veins. The result is a fall in blood pressure because of decreased peripheral resistance.

Terazosin is chemically related to prazosin and acts primarily by blocking α -1 adrenoceptors. It has similar pharmacodynamics effects to prazosin but a longer duration of action which enables once daily dosing. Terazosin has been shown to have neutral or possibly slightly beneficial effects on plasma lipids.

In animals, terazosin causes a decrease in blood pressure by decreasing total peripheral vascular resistance. Terazosin decreases blood pressure gradually within 15 minutes following oral administration.

In man, systolic and diastolic blood pressures are lowered in both the supine and standing positions. The effect is most pronounced on the diastolic blood pressure. These changes are usually not accompanied by reflex tachycardia. A greater blood pressure effect associated with peak plasma concentration (first few hours after dosing) appears somewhat more position dependent (greater in the erect position) than the effect of terazosin at 24 hours, and in the erect position there is also a 6-10 beat per minute increase in heart rate in the first few hours after dosing.

Studies suggest that α -1 adrenoceptor blockade is also useful in improving the urodynamics in patients with chronic bladder outlet obstruction such as in benign prostatic hyperplasia (BPH). The symptoms of BPH are caused mainly by the presence of an enlarged prostate and by the increased smooth muscle tone of the bladder outlet and the prostate, which is regulated by α -1 adrenergic receptors.

PHARMACOKINETICS:

Terazosin hydrochloride administered as terazosin tablets is essentially completely absorbed in man. Terazosin has been shown to undergo minimal hepatic first-pass metabolism and nearly all of the circulating dose is in the form of parent drug. The plasma levels peak about one hour after dosing, and then decline with a half-life of approximately 12 hours. The drug is highly bound to plasma proteins and binding is constant over the clinically observed concentration range. Approximately 10% of an orally administered dose is excreted as parent drug in the urine and approximately 20% is excreted in the faeces. The remainder is eliminated as metabolites. Overall, approximately 40% of the administered dose is excreted in the urine and approximately 60% in the faeces.

The pharmacokinetics of terazosin appears to be independent of renal function. This would obviate the need to adjust dosing regimens for patients with impaired renal function.

INDICATION:

Indicated for the relief of the manifestations of mild to moderate benign prostatic hyperplasia (BPH). Treatment should be stopped if patients have not responded after 3 months of therapy.

Terazosin is also indicated in the treatment of hypertension. It can be used alone or in combination with other antihypertensive agents such as diuretics or β -adrenergic blocking agents.

RECOMMENDED DOSAGE:

The dose of terazosin should be adjusted according to the patients individual response. The following is a guide to its administration.

Initial Dose: 1 mg at bedtime is the starting dose for all patients, and this dose should not be exceeded. This initial dosing regimen should be strictly observed to minimize the potential for severe hypotensive effects.

Subsequent Doses: Benign Prostatic Hyperplasia (BPH): The dose may be slowly increased to achieve the desired clinical response in BPH patients. The usual recommended dose range is 5 to 10 mg administered once a day. Treatment should commence with one tablet of 1 mg taken at bedtime for the first 4 days, then every morning for the next 3 days. For the next seven days, the dosage is one tablet of 2 mg every morning. In the third week, the dosage is increased to one tablet 5 mg every morning.

As from the fourth week, a maintenance dosage of 5-10 mg is taken daily each morning.

Urine flow rate measured approximately 24 hours after the last dose has shown that the beneficial effects in BPH persist for the recommended dosing interval.

Symptom improvements have been detected as early as two weeks after starting treatment with terazosin. Improvements in flow rate may be seen somewhat later.

Maximal benefit appears to occur after 3 to 6 months of treatment, but it appears that there is little to be gained by continuing treatment for more than 6 months if there has been no clinical response. Efficacy has been demonstrated in clinical studies of up to 18 months but data on longer term use is not yet available. If terazosin administration is discontinued for several days or longer, therapy should be reinstituted using the initial dosing regimen.

Hypertension: The dose may be slowly increased to achieve the desired blood pressure response. The usual recommended dose range is 1 mg to 5 mg administered once a day; however, some patients may benefit from doses as high as 20 mg per day. Doses over 20 mg do not appear to provide further blood pressure effect and doses over 40 mg have not been studied. Blood pressure should be monitored at the end of the dosing interval to be sure control is maintained throughout the interval. It may also be helpful to measure blood pressure 2-3 hours after dosing to see if the maximum and minimum responses are similar, and to evaluate symptoms such as dizziness or palpitations which can result from excessive hypotensive response. If the response is substantially diminished at 24 hours, an increase in dose or use of a twice daily regimen can be considered. If terazosin administration is discontinued for several days or longer, therapy should be reinstituted using the initial dosing regimen. In clinical trials, except for the initial dose, the dose was given in the morning.

Use with other Drugs: Caution should be observed when terazosin is administered concomitantly with other antihypertensive agents (e.g. calcium antagonists) to avoid the possibility of significant hypotension. When adding a diuretic or other antihypertensive agent, dosage reduction and retitration may be necessary. (See *Drug Interactions*).

There are no data available concerning the use of terazosin in combination with other agents used for treating benign prostatic hyperplasia, eg finasteride.

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS:

Patients known to be hypersensitive to terazosin hydrochloride or its analogues.

WARNINGS & PRECAUTIONS:

Warnings: Syncope and "First-dose" Effect: may occur when used in both hypertension and BPH. Terazosin, like other α -adrenergic blocking agents, can cause marked hypotension, especially postural hypotension, and syncope in association with the first dose or first few doses of therapy. Similar effect can be anticipated if therapy is interrupted for more than a few doses. To decrease the likelihood of syncope or excessive hypotension, treatment should always be initiated with a 1 mg dose of terazosin, given at bedtime. The 2 mg, 5 mg, and 10 mg tablets are not indicated as initial therapy.

Dosage should then be increased slowly and additional antihypertensive agent should be added with caution. The patient should be cautioned to avoid situations where injury could result should syncope occur during initiation of therapy.

If syncope occurs, the patient should be placed in the recumbent position and treated supportively as necessary. There is evidence that the orthostatic effect of terazosin is greater, even in chronic use, shortly after using.

Precautions: General Prostatic Cancer: Carcinoma of the prostate and BPH causes many of the same symptoms. These 2 diseases frequently coexist. Therefore, patients thought to have BPH should be examined prior to starting terazosin therapy to rule out the presence of carcinoma of the prostate.

Orthostatic Hypotension: While syncope is the most severe orthostatic effect of terazosin (see Warning), other symptoms of lowered blood pressure, such as dizziness, lightheadedness, and palpitations, are more common. Patients with occupations in which such events represent potential problems should be treated with particular caution.

Information for patient: Patient should be made aware of the possibility of syncopal and orthostatic symptoms, especially at the initiation of the therapy, and to avoid driving or hazardous tasks for 12 hours after the first dose, after dosage increase, and after interruption of therapy when treatment is resumed. They should be cautioned to avoid situation where injury could result should syncope occur during initiation of terazosin therapy. They should also be advised of the need to sit or lie down when symptoms of lowered blood pressure occur, although these symptoms are not always orthostatic, and to be careful when rising from a sitting or lying positions. If dizziness, lightheadedness, or palpitations are bothersome they should be reported to the physician, so that dose adjustment can be considered. Patient should also be told that drowsiness or somnolence can occur with terazosin, requiring caution in people who must drive or operate heavy machinery.

Paediatric Use: Safety and effectiveness in children have not been determined.

INTERACTION WITH OTHER MEDICAMENTS:

Hypertension: In controlled trials, terazosin has been added to diuretics, and several β -adrenergic blockers; no unexpected interactions were observed. Terazosin has also been used in patients on a variety of concomitant therapies: while these were not formal interactions studies, no interactions were observed. Terazosin has been used concomitantly in at least 50 patients on the following drugs or drug classes:

- 1) analgesic/anti-inflammatory (e.g., paracetamol, aspirin, codeine, ibuprofen, indomethacin);
- 2) antibiotics (e.g., erythromycin, trimethoprim and sulphamethoxazole);
- 3) anticholinergic/sympathomimetics (e.g. phenylephrine hydrochloride, phenylpropanolamine hydrochloride, pseudoephedrine hydrochloride);
- 4) antigout (e.g., allopurinol);
- 5) antihistamines (e.g., chlorpheniramine);
- 6) cardiovascular agents (e.g., atenolol, hydrochlorothiazide, methyclothiazide, propranolol);
- 7) corticosteroids;

- 8) gastrointestinal agents (e.g antacids);
- 9) hypoglycemics;
- 10) sedatives and tranquilizer (e.g., diazepam).

Benign Prostatic Hyperplasia (BPH): In clinical trials in BPH patients the number reporting dizziness or other dizziness -related adverse events appears to be greater in those patients receiving terazosin and ACE inhibitors or diuretics and the number reporting asthenia was greater in those receiving terazosin and NSAID, than in the total population of terazosin patients from double-blind, placebo-controlled studies. No interactions were observed in patients treated concurrently with theophylline, anti-anginal agents or oral hypoglycaemic agents.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Terazosin hydrochloride is not recommended during pregnancy unless the potential benefit justifies the potential risk to mother and foetus.

Use in Lactation: It is not known whether terazosin is excreted in breast milk. Because many drugs are excreted in breast milk, caution should be exercised when terazosin is administered to a nursing woman.

SIDE EFFECTS:

Most adverse effects encountered during clinical studies have been of mild to moderate severity. The most frequent effects are dizziness, headache, asthenia, nasal congestion and cold symptoms. Other adverse effects include somnolence, nausea, blurred vision, tachycardia, palpitations, chest pain, dyspnoea, and syncope. First dose hypotension was largely avoided by limiting the initial dose of terazosin to 1 mg and giving it at bedtime, but a syncopal episode was reported in about 1% of patients at some time during therapy. Terazosin has no effect on renal function in patients with normal, or moderately or severely impaired renal function.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Should overdosage of terazosin lead to hypotension, support of the cardiovascular system is of first importance. Restoration of blood pressure and normalization of heart rate may be accomplished by keeping the patient in the supine position. If this measure is inadequate, shock should first be treated with volume expanders. If necessary, vasopressor should then be used and renal function should be monitored and supported as needed. Laboratory data indicate that terazosin is highly protein-bound; therefore, dialysis may not be benefit.

STORAGE CONDITIONS:

Store below 30°C. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

RALSIN TABLET 2MG: 10 x 10's in blister pack.

RALSIN TABLET 5MG: 10 x 10's in blister pack.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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